

The Chain Rule

Theorem. Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open, let $g : U \rightarrow V$ be differentiable at $a \in U$ and let $f : V \rightarrow \mathbb{R}^p$ be differentiable at $b = g(a)$. Then $f \circ g$ is differentiable at a and $D(f \circ g)(a) = Df(b) \circ Dg(a)$.

Proof. Write $L_g = Dg(a)$ and $L_f = Df(b)$. By differentiability of g at a we have $g(a+h) = g(a) + L_g(h) + E_g(h)$ where $\lim_{h \rightarrow 0} |E_g(h)|/|h| = 0$. Let $k(h) = L_g(h) + E_g(h)$. Then $f(g(a+h)) = f(b + k(h)) = f(b) + L_f(k(h)) + E_f(k(h))$, so that $f(g(a+h)) - f(g(a)) - L_f(L_g(h)) = L_f(E_g(h)) + E_f(k(h))$. Since L_f is linear it is bounded, say $|L_f(v)| \leq \|L_f\|_{\text{op}}|v|$ for all $v \in \mathbb{R}^m$, hence $|L_f(E_g(h))|/|h| \leq \|L_f\|_{\text{op}}|E_g(h)|/|h| \rightarrow 0$ as $h \rightarrow 0$. For the second term, $|k(h)| \leq \|L_g\|_{\text{op}}|h| + |E_g(h)| \leq (\|L_g\|_{\text{op}} + 1)|h|$ for $|h|$ small, and $E_f(k)/|k| \rightarrow 0$ as $k \rightarrow 0$, so $|E_f(k(h))|/|h| \leq (\|L_g\|_{\text{op}} + 1)|E_f(k(h))|/|k(h)| \rightarrow 0$ as well (with the convention $E_f(0)/|0| = 0$). Therefore $f \circ g$ is differentiable at a with derivative $L_f \circ L_g$.

Sequences and series

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers with $a_n \rightarrow a$ and let $(b_n)_{n \in \mathbb{N}}$ satisfy $|b_n - b| \leq 2^{-n}$ for every $n \geq n_0$. Then $a_n + b_n \rightarrow a + b$, $a_n b_n \rightarrow ab$ and, provided $b \neq 0$ and $b_n \neq 0$ for all n , $a_n/b_n \rightarrow a/b$; moreover $\sup_{n \geq N} |a_n - a| \rightarrow 0$ and $\inf_{n \geq N} a_n \rightarrow a$ as $N \rightarrow \infty$, and if $a_n \leq c_n \leq b_n$ for all $n \geq n_1$ with $b_n \rightarrow a$ then $c_n \rightarrow a$ too. If in addition $\sum_{n=1}^{\infty} |a_n - a| < \infty$ then $\sum_{n=1}^N (a_n - a) \rightarrow s$ for some $s \in \mathbb{R}$ with $|s| \leq \sum_{n=1}^{\infty} |a_n - a|$, and $\sum_{n=1}^N a_n - Na \rightarrow s$ as $N \rightarrow \infty$.

Linear maps

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear with matrix $A = (a_{ij})$ so that $T(e_j) = \sum_{i=1}^m a_{ij}e_i$ for $1 \leq j \leq n$. For $x = \sum_j x_j e_j$ we obtain $T(x) = \sum_{i=1}^m (\sum_{j=1}^n a_{ij}x_j)e_i$ and hence $|T(x)| \leq \sum_{i,j} |a_{ij}||x_j| \leq (\sum_{i,j} |a_{ij}|)|x|$, which shows $\|T\|_{\text{op}} \leq \sum_{i,j} |a_{ij}| < \infty$; conversely $\|T\|_{\text{op}} \geq |T(e_j)| = |(a_{1j}, \dots, a_{mj})| \geq |a_{ij}|$ for every i and j , so $\max_{i,j} |a_{ij}| \leq \|T\|_{\text{op}} \leq mn \max_{i,j} |a_{ij}|$. The kernel $\ker T = \{x \in \mathbb{R}^n : T(x) = 0\}$ and the image $T(\mathbb{R}^n) = \{T(x) : x \in \mathbb{R}^n\}$ satisfy $\dim \ker T + \dim T(\mathbb{R}^n) = n$, and T is injective if and only if $\ker T = \{0\}$, if and only if $|T(x)| \geq c|x|$ for some $c > 0$ and all $x \in \mathbb{R}^n$.

Number theory

For $a, b \in \mathbb{Z}$ not both zero let $d = \gcd(a, b)$. Then there exist $u, v \in \mathbb{Z}$ with $au + bv = d$, and $d \mid c$ whenever $c \in \mathbb{Z}$ satisfies $c = ax + by$ for some $x, y \in \mathbb{Z}$; in particular $\{ax + by : x, y \in \mathbb{Z}\} = d\mathbb{Z}$. If p is prime and $p \mid ab$ then $p \mid a$ or $p \mid b$, because otherwise $\gcd(p, a) = \gcd(p, b) = 1$ gives $1 = pu_1 + av_1 = pu_2 + bv_2$ and multiplying yields $1 = p(pu_1u_2 + au_2v_1 + bu_1v_2) + abv_1v_2 \equiv 0 \pmod{p}$, a contradiction. Consequently every $n \geq 2$ factors as $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$ with $p_1 < p_2 < \cdots < p_k$ prime and $e_i \geq 1$, uniquely, and $\tau(n) = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1)$, $\sigma(n) = \prod_{i=1}^k (p_i^{e_i+1} - 1)/(p_i - 1)$ and $\varphi(n) = n \prod_{i=1}^k (1 - 1/p_i)$.

Continuity and compactness

Suppose $K \subseteq \mathbb{R}^n$ is compact and $f : K \rightarrow \mathbb{R}$ is continuous. Then f is bounded and attains its bounds: there are $x_0, x_1 \in K$ with $f(x_0) \leq f(x) \leq f(x_1)$ for all $x \in K$. Indeed, choose $x_k \in K$ with $f(x_k) \rightarrow \sup_K f \in (-\infty, +\infty]$; by compactness a subsequence $x_{k_j} \rightarrow x_1 \in K$, and continuity gives $f(x_1) = \lim_{j \rightarrow \infty} f(x_{k_j}) = \sup_K f < \infty$. Moreover f is uniformly continuous: for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $x, y \in K$ and $|x - y| < \delta$, since otherwise we would find $x_k, y_k \in K$ with $|x_k - y_k| < 1/k$ but $|f(x_k) - f(y_k)| \geq \varepsilon$, and passing to a subsequence $x_{k_j} \rightarrow z, y_{k_j} \rightarrow z$ would contradict $f(x_{k_j}) - f(y_{k_j}) \rightarrow f(z) - f(z) = 0$.

Inequalities

For $x, y \geq 0$ and $p, q > 1$ with $1/p + 1/q = 1$ we have $xy \leq x^p/p + y^q/q$ (Young), hence for $u, v \in \mathbb{R}^n$ the estimate $\sum_{i=1}^n |u_i v_i| \leq (\sum_{i=1}^n |u_i|^p)^{1/p} (\sum_{i=1}^n |v_i|^q)^{1/q}$ (Hölder), and with $p = q = 2$ the Cauchy–Schwarz inequality $|\langle u, v \rangle| \leq |u| |v|$, with equality if and only if u and v are linearly dependent. From it, $|u + v|^2 = |u|^2 + 2\langle u, v \rangle + |v|^2 \leq |u|^2 + 2|u||v| + |v|^2 = (|u| + |v|)^2$, which is the triangle inequality $|u + v| \leq |u| + |v|$, and also $||u| - |v|| \leq |u - v|$, so that $x \mapsto |x|$ is 1-Lipschitz and in particular continuous on $\mathbb{R}^n$.